

ME350 FA 2025 Section 002

Lab #3

Monday 12:30 PM - 2:30 PM

Haoran Shi

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Agenda

- Reminders
- Manufacturing Plan Approval Process
- Joint Design Examples
- Key Points for Lab 3
- Self-paced Lab



Reminders

- Lab Assignment #3 (Individual ADAMS Model, CAD) due Sep. 22nd, 12:30 pm
- HW #2 due Thurs, Sep. 18th at 1:30pm on Gradescope
- Gate Review #1 is scheduled for **outside of lab** with your lab instructor by the day before Lab 4
- Drawings and Manufacturing Plans **due with DR1**
- Design Report #1 due:
 - Wednesday (9/26) - Monday Labs



Gate Review #1

Set up outside of lab with lab instructor

Announcement to be sent next week with numerous 30 min time slots

Requirements are in the “Lab Assignment Summary” document

Please bring to Gate Review #1:

1. Slides with:
 1. Link lengths, transmission angle deviations for positions 1, 3, and 4
 2. Joint design (sectional view of your 3D CAD model joints)
 3. ADAMS Model results
2. 3D SolidWorks Model with:
 1. Detailed coupler design (with flashlight mounting)
 2. All linkage components (links, ground plate, coupler, etc.)
 3. Hard stops (used to limit linkage motion)
 4. Spacers or any other components used for mounting to the playing field
 5. All fasteners, spacers, and other components used in joints
 6. Your linkage assembly on your playing field model. The linkage assembly should be mated in a way that allows the linkage to move through all 4 positions.

For your slides, just have one member of your team bring them on a laptop

For your CAD, have one member of your team bring it on their laptop if they have SolidWorks locally downloaded, or bring it on a flashdrive, or email it to the instructor before your GR1 slot.

Hard Stops, Spacers, and Motor

- Hard stops
 - They can be made from aluminum
 - **Think about adjustability of your hard stops...**
 - Can put rubber around them to soften the blow
 - You can mount these with a screw and a nut
- Mounting to Playing Field
 - The playing field has $\frac{3}{8}$ "-16 threads and $\frac{1}{4}$ " dowel pin holes
 - We have screws and dowels available
 - If you choose to use dowel pins:
 - You should have **at least two** screws connecting your mechanism to the playing field (for rigidity)
- Motor mounting
 - **Leave space to mount the motor with a bracket**
 - Motor CAD is available in the ME350 Hardware Files

Calendar

		Sep 14	Sep 15	Sep 16	Sep 17	Sep 18	Sep 19	Sep 20
4	Lecture			Lecture 7 - DC Motors (DB)		Lecture 8 - DC Motors, Transmissions (DB)		
	Homework					- Homework 2 Due (Synthesis, ICs, MA) - Homework 3 Released (Motors)		
	Lab		Lab 3 - Linkage CAD Assembly Tutorial, Fasteners and bearings discussion / Individual ADAMS Lab Assignment #2 Due - Individual ADAMS Tutorial Graph + Individual Solidworks Links Team Expectations Agreement Due DR1 Section 1 Due			Lab 4 Manufacturing Review and Project Work Lab Assignment #3 Due - Individual ADAMS Graphs using Individual Solidworks Links	Lab 4 Manufacturing Review and Project Work Lab Assignment #3 Due - Individual ADAMS Graphs using Individual Solidworks Links	
	Project Timeline		<--Gate Review 1 Out of Lab (Scheduled by Team)					
		Sep 21	Sep 22	Sep 23	Sep 24	Sep 25	Sep 26	Sep 27
5	Lecture			Lecture 9 - Inertia Matching, Planetary Gears (MU)		Lecture 10 - Review for Midterm Exam (MU and DB)		
	Homework					Homework 3 Due (Motors)		
	Lab		Lab 4 Manufacturing Review and Project Work Lab Assignment #3 Due - Individual ADAMS Graphs using Individual Solidworks Links			Lab 5 - Transmission Selection Tutorial	Lab 5 - Transmission Selection Tutorial	
	Project Timeline		DR 1 DUE TODAY AT MIDNIGHT	DR 1 DUE TODAY AT MIDNIGHT			DR 1 DUE TODAY AT MIDNIGHT	
			Gate Review 1 Out of Lab-->					

Manufacturing Plan Approval Process



Manufacturing Review

- In lab 4.
- Goal: Discuss manufacturability with one other team + GSI/IA
 - Not a formal presentation: no need to dress up
 - **Attendance is mandatory** for all teams for all presentations
 - Every team gets 10 min to present (shorter is better) and 5 min for Q & A

Time Slot	Presenting Team (reviewing team)
12.45 pm - 1.00 pm	Team #1 (#7)
1.00 pm - 1.15 pm	Team #2 (#1)
1.15 pm - 1.30 pm	Team #3 (#2)
1.30 pm - 1.45 pm	Team #4 (#3)
1.45 pm - 2.00 pm	Team #5 (#4)
2.00 pm - 2.15 pm	Team #6 (#5)
2.15 pm - 2.30 pm	Team #7 (#6)



Manufacturing Review

As design team prepare slides with at least the following

- Overall design (3D CAD from GR1)
 - Coupler and flashlight mounting
 - Joint design
 - Hard stops
- Engineering drawings
- Anything additional comments/concerns
 - special features
 - manufacturing difficulties
 - tight tolerances
 - etc
- **Max 10 min to present!**



Manufacturing Approval Process

NEW DRAWINGS (NOT for drawing revisions)

- First, submit your draft to the form
- [Fill out the form](#)
 - Please use the drawing template found [here](#) on main Canvas
 - MECHENG 350 001 > Files > Project Resources > ME350 Drawing Template.SLDDRW
 - Each drawing must be **checked** by another team member before submission
- After you fill out this form, your GSI will be automatically emailed. Then either:
 - Drawing approved by GSI and is then sent to the machine shop for final approval
 - Your drawing is rejected



Rejected Drawings

If your drawing is rejected by your GSI, you will get an email that looks like this.

1. Click on the drawing to see your GSI's comments
2. Click on the link to submit your revised drawing

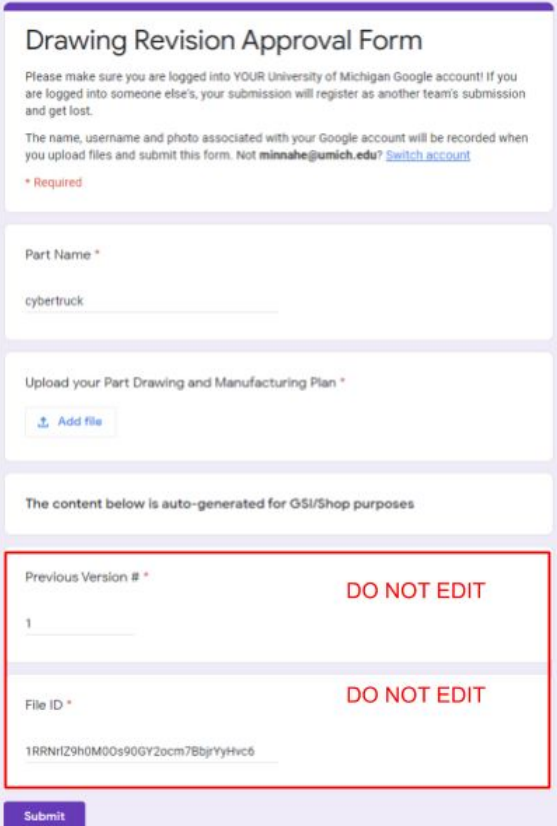


Submitting Revised Drawings

After clicking the link, the form will look like this and is **automatically pre-filled out for you!**

- All you need to do is upload your **revised** drawing and manufacturing plan
- **DO NOT EDIT the bottom two boxes**
 - This may break the auto-generated links, and your drawing may be lost!

You will go through the same process for drawings rejected by the machine shop, but your GSI **does not** need to



The screenshot shows a web form titled "Drawing Revision Approval Form". It includes instructions about logging into a University of Michigan Google account. The form has several input fields: "Part Name" (pre-filled with "cybertruck"), "Upload your Part Drawing and Manufacturing Plan" (with an "Add file" button), "Previous Version #" (pre-filled with "1"), and "File ID" (pre-filled with a long alphanumeric string). The last two fields are highlighted with a red border and labeled "DO NOT EDIT". A "Submit" button is at the bottom.

Drawing Revision Approval Form

Please make sure you are logged into YOUR University of Michigan Google account! If you are logged into someone else's, your submission will register as another team's submission and get lost.

The name, username and photo associated with your Google account will be recorded when you upload files and submit this form. Not minnahe@umich.edu? [Switch account](#)

* Required

Part Name *

cybertruck

Upload your Part Drawing and Manufacturing Plan *

[Add file](#)

The content below is auto-generated for GSI/Shop purposes

Previous Version # *

1

DO NOT EDIT

File ID *

1RRNrI29h0M00s90GY2ocm78bpYyHvc6

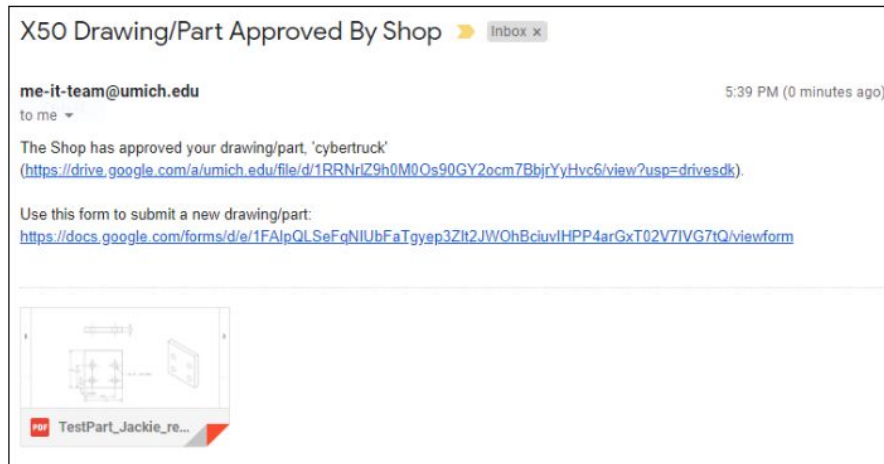
DO NOT EDIT

Submit

Approved Drawings

You will be email notified when your drawing is approved by both the GSI and the machine shop

- Make sure both signatures (your GSI and the machine shop) are in the title box



	NAME	DATE	ME 250 TEAM 92		
DRAWN	WAIERMA	11/5/19	TITLE: MOTOR BRACKET SPACER		
CHECKED	LIAMM	11/5/19			
GSI/IA APPR.	schmiedj 11/10				
SHOP APPR.	dwirkner 11/12/19				
INSPECTED			SIZE	DWG. NO.	REV
COMMENTS:			A	P92_006	3
			SCALE: 4:1	QUANTITY: 1	SHEET 1 OF 3

Joint Design

Joint Design Components

Table 4: Available in the machine shop or X50 assembly area (1510 GGBL)

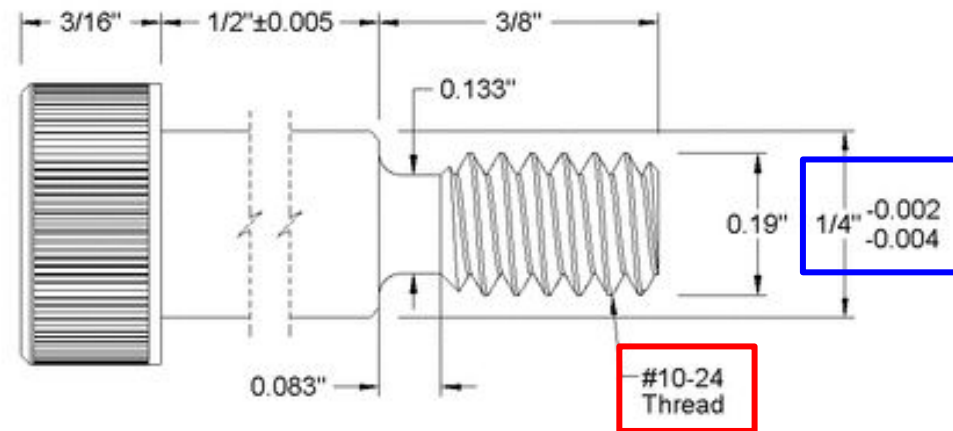
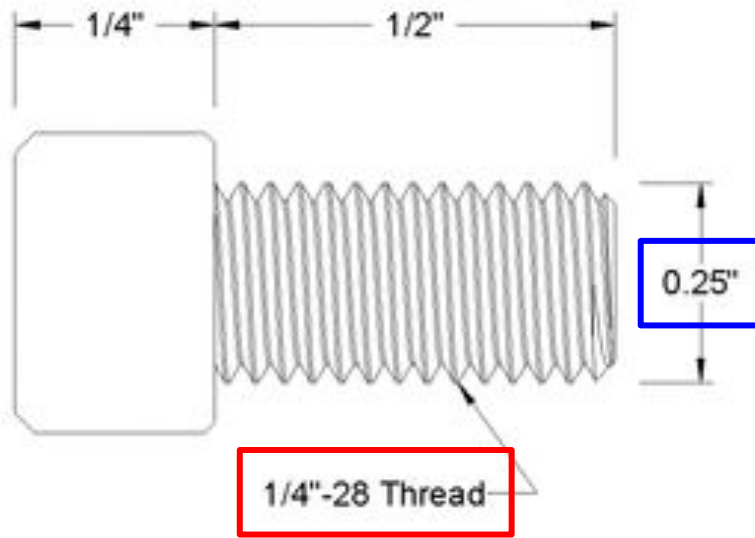
Item Description	Part #
Nylon Unthreaded Spacers, 5/8" OD, 1/2" Length, for 1/4" ID	94639A694
Shoulder Screw, 1/4" Diameter x 1/2" Shoulder, 10-24 Thread <i>(Other lengths of shoulder screws will also be available)</i>	91259A537
Oil-Embedded Thrust Bearing (Washer) for 1/4" Shaft Dia., 5/8" OD, 1/16" Thk.	5906K531
Nylon Locknut, 10-24 Thread Size	90631A011
Socket Head Cap Screw, 3/8"-16 Thread, 2" Length	91251A632
Socket Head Cap Screw, 3/8"-16 Thread, 2-1/2" Length	91251A634



<https://www.mcmaster.com/shoulders>

Various lengths of shoulder screws available – check X50 assembly room!

Joint Design Components



- Socket head cap screw VS. socket head shoulder screw
- Both are nominally 1/4" diameter!

Joint Design Components

Table 4: Available in the machine shop or X50 assembly area (1510 GGBL)

Item Description	Part #
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www.mcmaster.com/5906k531

Table 3: Kit Items

Item Description	Part #	Supplier
Aluminum Plate, 1/4" x 12" x 18"	21430300	Alro Steel
Aluminum Angle, 2.5" x 2" x 1/4" thick, 6" long	Custom	Alro Steel
Aluminum round stock, 1" Diameter, 12" Long	Custom	Alro Steel
SAE 841 Sleeve Bushing for 1/4" Shaft Dia., 1/4" Lg.	6391K126	McMaster-Carr
Needle Thrust Bearing, Bore .250	4XFL8	Grainger

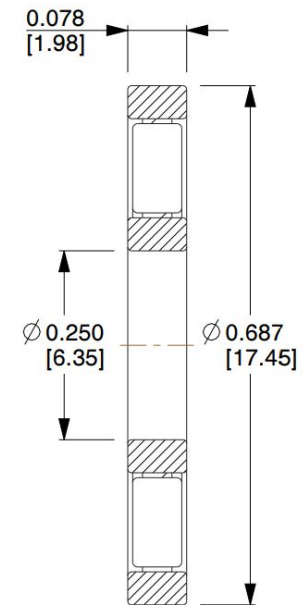
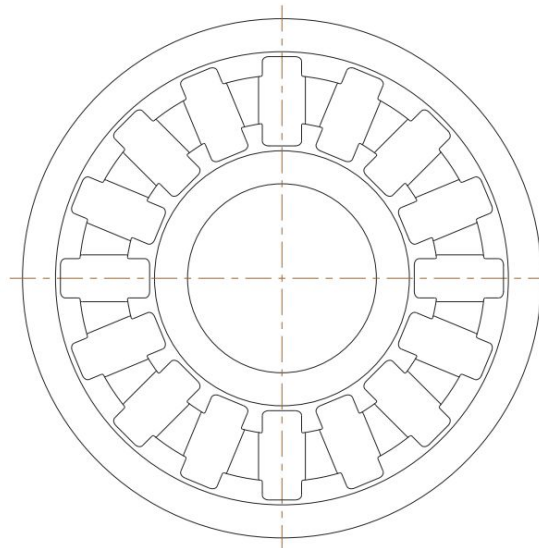


www.mcmaster.com/6391k126

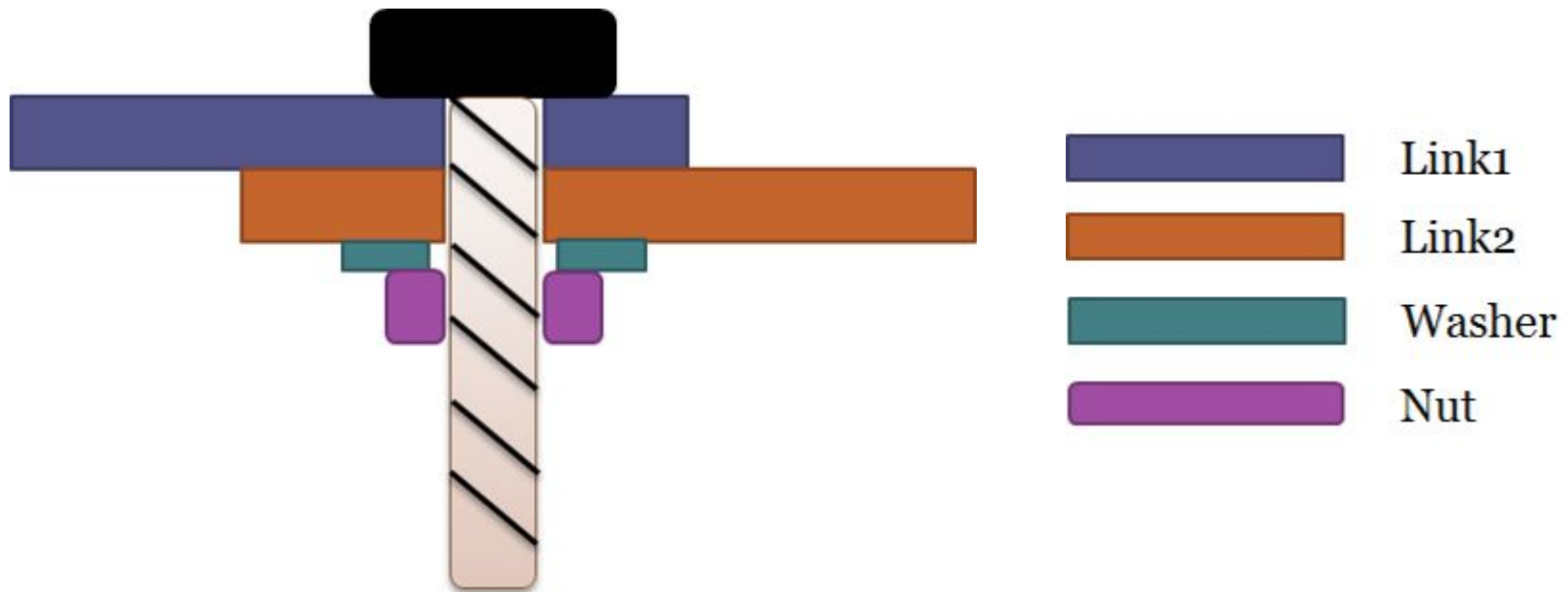
Joint Design Components

Table 3: Kit Items

Item Description	Part #	Supplier	Qty per team
Aluminum Plate, 1/4" x 12" x 18"	21430300	Alro Steel	1
Aluminum Angle, 2.5" x 2" x 1/4" thick, 6" long	Custom	Alro Steel	1
Aluminum round stock, 1" Diameter, 12" Long	Custom	Alro Steel	1
SAE 841 Sleeve Bushing for 1/4" Shaft Dia., 1/4" Lg.	6391K126	McMaster-Carr	10
Needle Thrust Bearing, Bore .250	4XFL8	Grainger	4

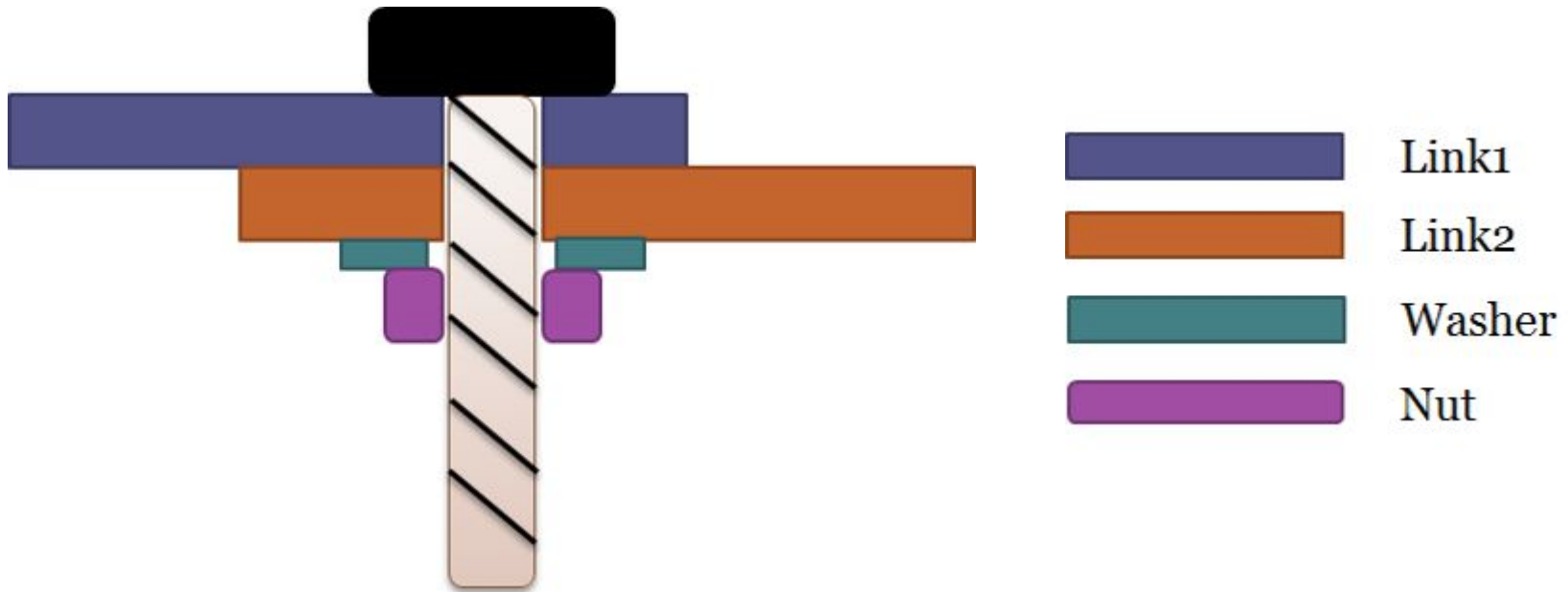


Joint Design Example



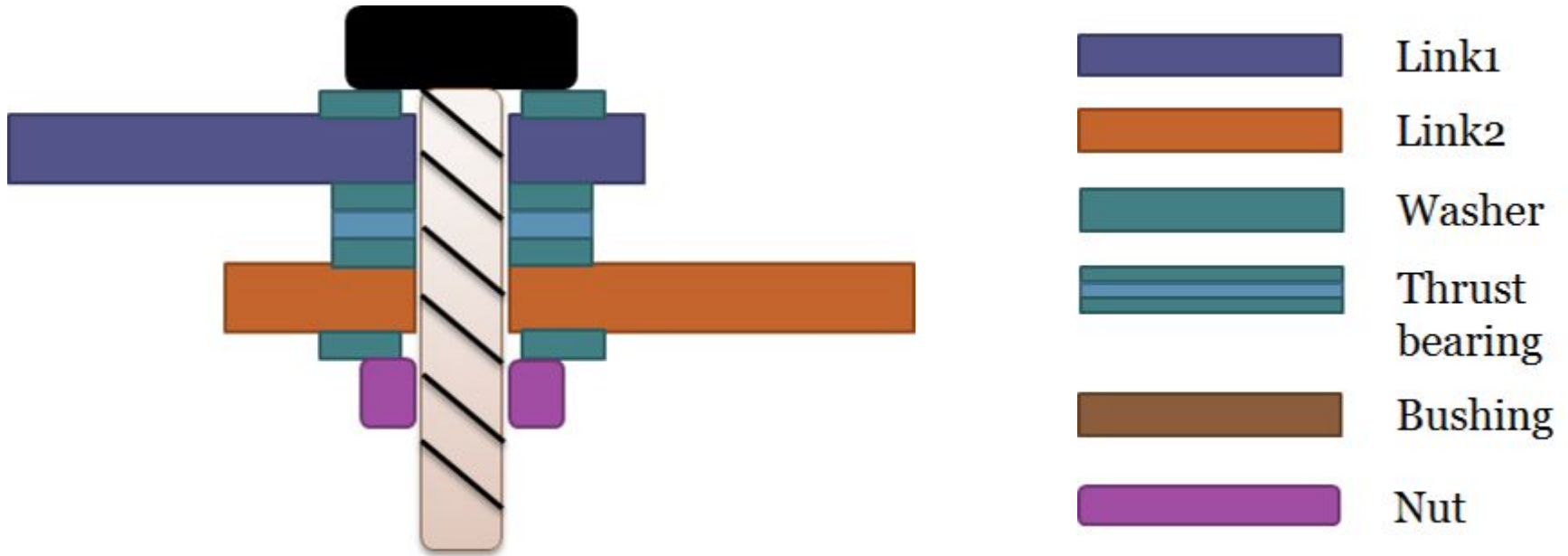
Discuss some pros and cons of this design

Joint Design Example



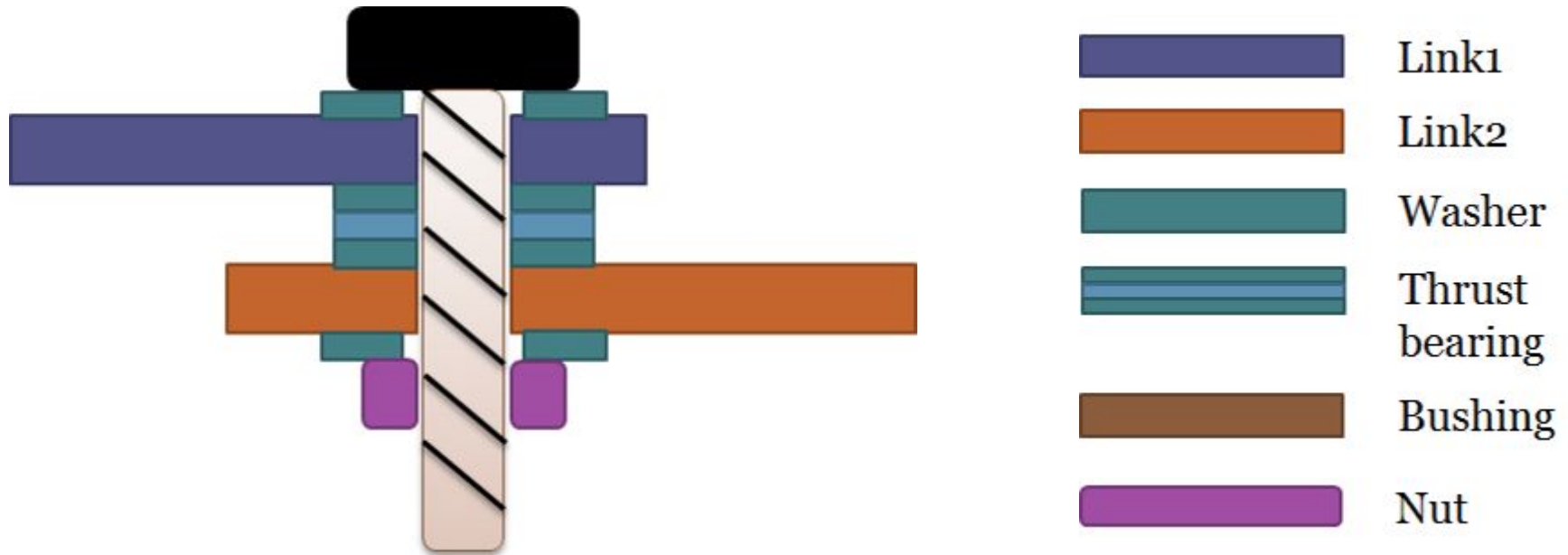
- ✓ Links are held together
- ✗ High friction between links
- ✗ Links are both tapped, inhibiting rotation between them
- ✗ Bolt sticks out very far from links
- ✗ Nut can be overtightened, inhibiting rotation

Joint Design Example



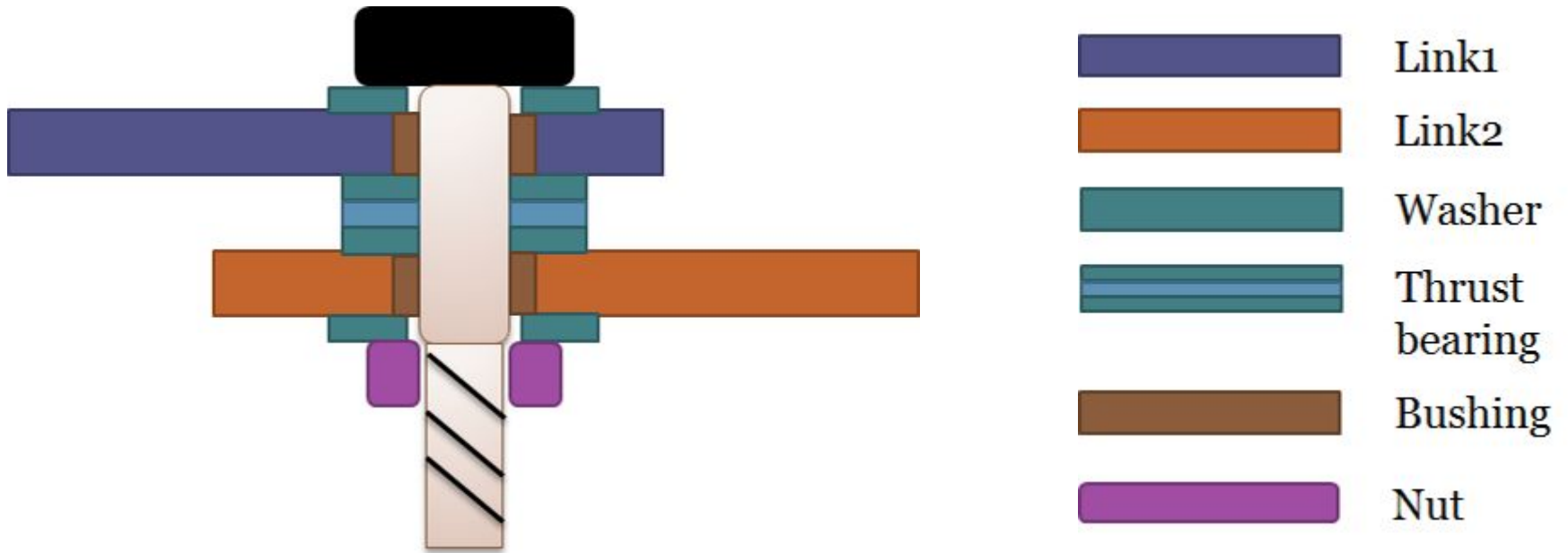
Discuss some pros and cons of this design

Joint Design Example



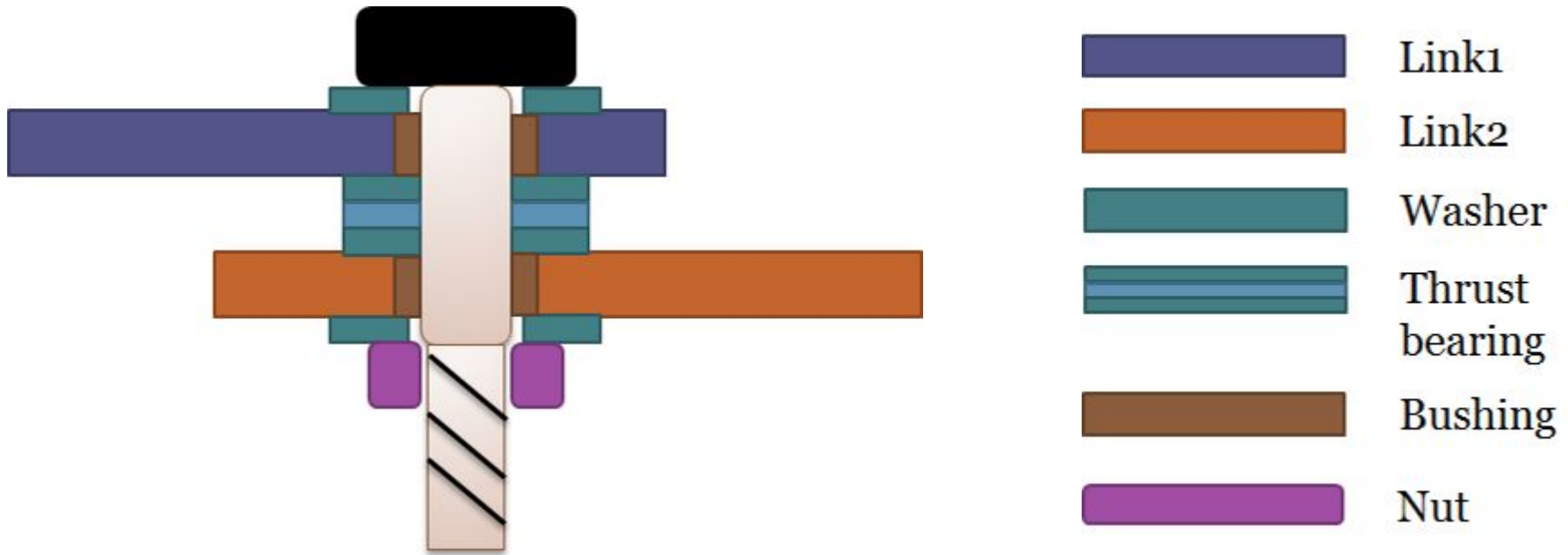
- ✓ Washers and thrust bearing used to reduce friction
- ✗ Links are both tapped, inhibiting rotation between them
- ✗ Washers should go above and below thrust bearing to distribute load
- ✗ Nut can be overtightened, inhibiting rotation

Joint Design Example



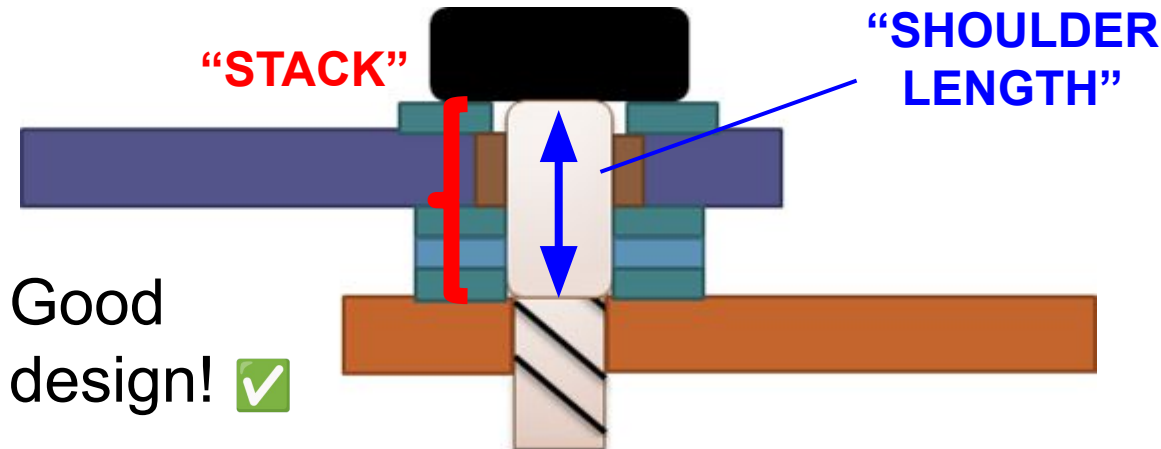
Discuss some pros and cons of this design

Joint Design Example



- ✓ Bushings and shoulder bolt allow links to rotate
- ✗ The point where the shoulder bolt tapers and becomes threaded should rest on a link, not go through it
- ✗ Washers should go above and below thrust bearing to distribute load
- ✗ Nut can be overtightened, inhibiting rotation

Joint Design Example



	Link1
	Link2
	Washer
	Thrust bearing
	Bushing
	Nut

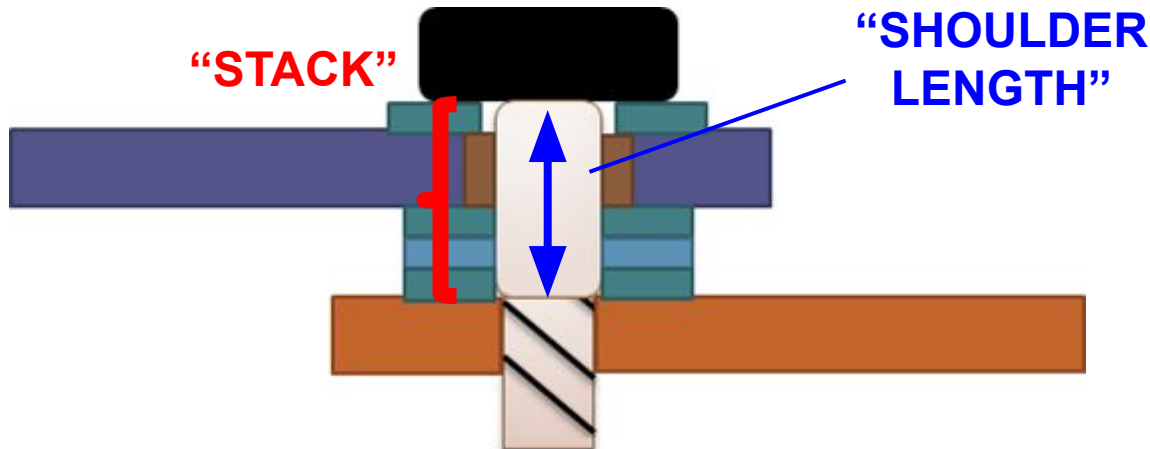
Currently shown: **SHOULDER LENGTH == STACK**

What happens if **SHOULDER LENGTH > STACK** when you tighten the screw?

What happens if **SHOULDER LENGTH < STACK** when you tighten the screw?

What happens if **SHOULDER LENGTH < STACK** when you slightly tighten the screw?

Joint Design Example



	Link1
	Link2
	Washer
	Thrust bearing
	Bushing
	Nut

Currently shown: **SHOULDER LENGTH == STACK**

What happens if **SHOULDER LENGTH > STACK** when you tighten the screw? *Joint is loose -- it has "end-play"!*

What happens if **SHOULDER LENGTH < STACK** when you tighten the screw? *Joint is tightened and can't rotate!*

What happens if **SHOULDER LENGTH < STACK** when you slightly tighten the screw? *Screw will undo itself!*

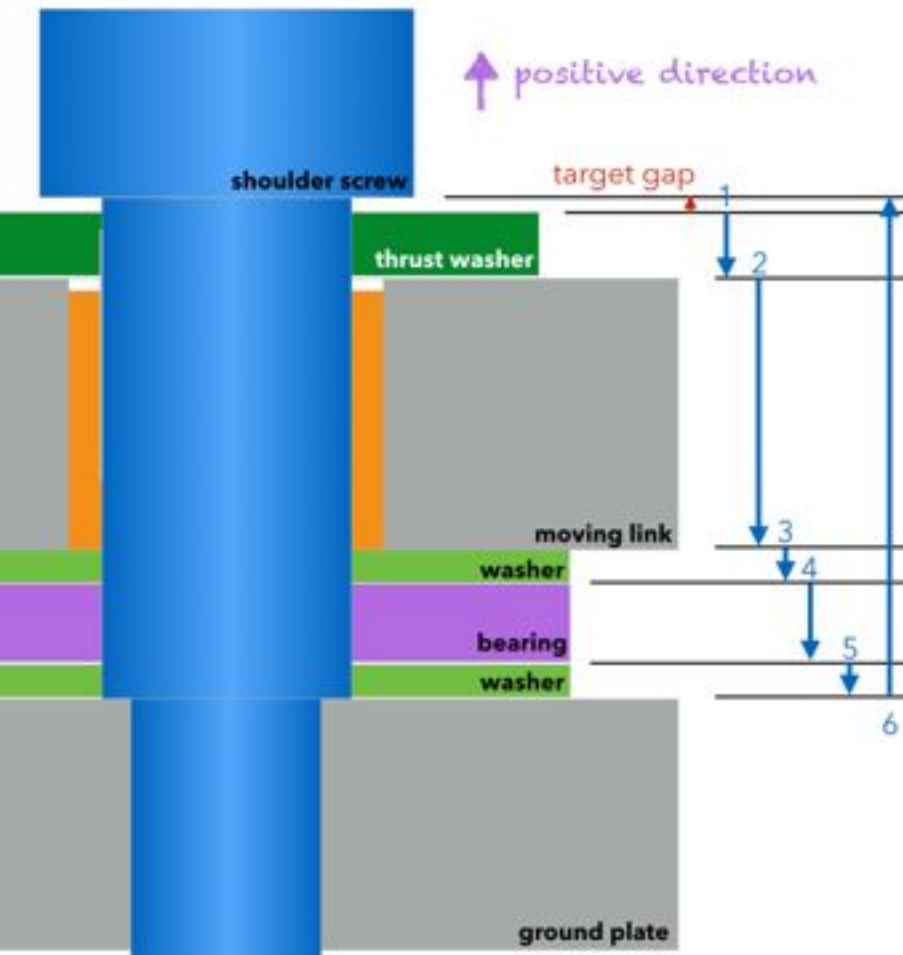
Joint Design Example

This is an
example of a
good design!



Joint Design: Detail

By analyzing the joint stack - you can easily determine slack/interference



Tolerance Analysis: Joint Wiggle vs Over-Tighten

step	item	dir	nom	- tol	+ tol	min	nom	max
1	thrust washer	-1	0.0625	0.0025	0.0025	0.0025	-0.0625	0.0025
2	plate	-1	0.25	0.0012	0.0012	0.0012	-0.25	0.0012
3	washer	-1	0.032	0.002	0	0	-0.032	0.002
4	bearing	-1	0.078125	0.0004	0	0	-0.0781	0.0004
5	washer	-1	0.032	0.002	0	0	-0.032	0.002
6	shoulder	1	0.500	0.005	0.005	0.005	0.5	0.005
Total gap						0.0087		0.0131
						0.037	0.045	0.058

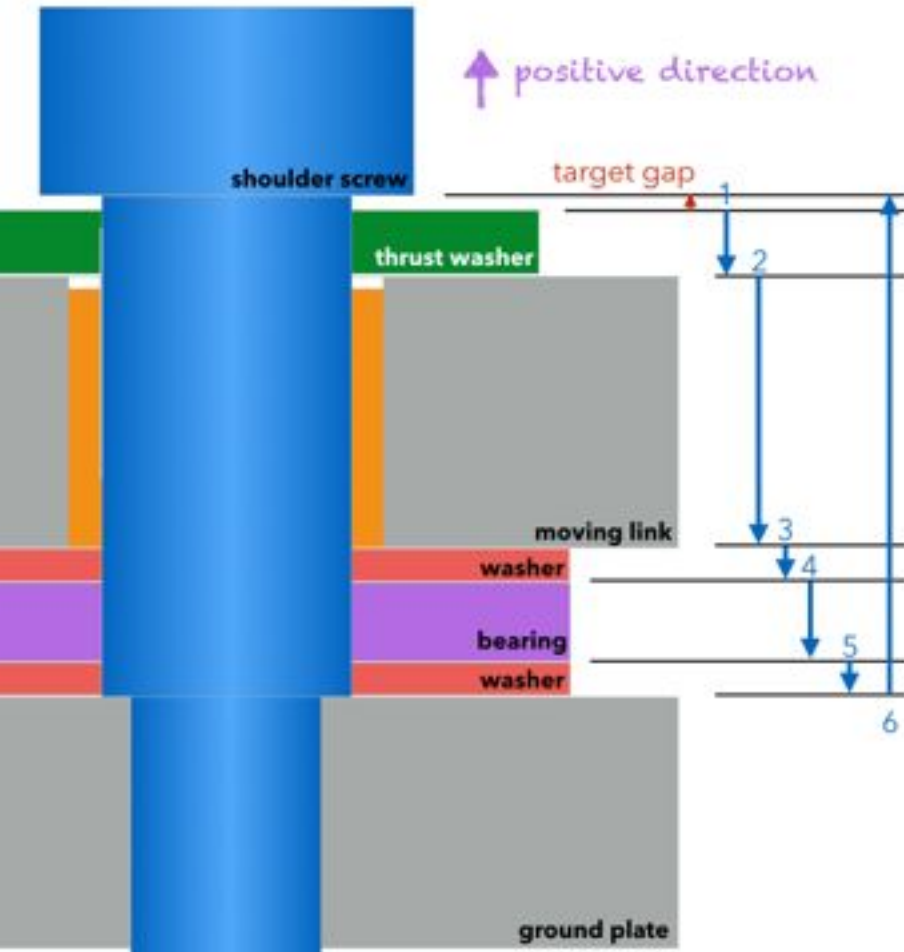
Result: in both nominal and worst-case tolerance conditions, the joint will have a gap (won't be over-tightened).

ASSUMPTIONS:

1. Shoulder screw is fully engaged into lower link.
2. Bushing is pressed flat against one side of plate (share a common face).
3. Plate is thicker than bushing (src: tolerance on plate is looser than the tolerance on bushing, same nominal thickness).

Joint Design: Detail

Be careful, your tolerance analysis will change if you use different parts!



Tolerance Analysis: Joint Wiggle vs Over-Tighten

step	item	dir	nom	- tol	+ tol	min	nom	max
1	thrust washer	-1	0.0625	0.0025	0.0025	0.0025	-0.0625	0.0025
2	plate	-1	0.25	0.0012	0.0012	0.0012	-0.25	0.0012
3	washer	-1	0.06	0.01	0.01	0.01	-0.06	0.01
4	bearing	-1	0.078125	0.0004	0	0	-0.0781	0.0004
5	washer	-1	0.06	0.01	0.01	0.01	-0.06	0.01
6	shoulder	1	0.500	0.005	0.005	0.005	0.5	0.005
Total gap						0.0287		0.0291
						-0.039	-0.011	0.018

Result: By swapping with an off-the-shelf washer, the joint is likely to be over-compressed once the shoulder screw is tightened.

ASSUMPTIONS:

1. Shoulder screw is fully engaged into lower link.
2. Bushing is pressed flat against one side of plate (share a common face).
3. Plate is thicker than bushing (src: tolerance on plate is looser than the tolerance on bushing, same nominal thickness).

Joint Design Options

- You DO NOT have to limit yourself to kit materials

- Check out:
 - McMaster Carr
 - Grainger
 - Existing joint designs



Vehicle Light
IP67 Rated, 5" Overall Diameter
www.mcmaster.com/1635k56

NOT ALLOWED

You have up to \$100 to spend on your linkage. Make sure you reserve enough budget for your transmission components.

Drafting Tips

Tolerance

- Commonly used system of general tolerance
- Unless otherwise specified, all:
 - .XX = +/- .010 inch
 - .XXX = +/- .005 inch
- In general, any dimension in the inch system should be 2-place precision, unless it must be a tighter tolerance due to mating
- Use 3-place precision between two bolt holes
- Use +/- .001 inch or better between two dowel holes



Tolerance



Air Bearing Lathe achieves 0.00004"
resolution



Horizontal Bandsaw achieves
~0.0625" resolution maximum

Why use manufacturing processes/tools that cannot hold high tolerances?

Lathe holds high tolerance, but takes a long time. Time eventually has a cost.

\$\$\$

Bandsaw holds lower tolerances, but is fast and easy to operate.

\$

Fits

- **Clearance fits** for when it may be desirable for the shaft to rotate or slide freely within the hole, this is usually referred to as a “sliding fit” or a “free fit”.
(loose)
- **Transition (close) fits** for when it is desirable that the shaft be held securely, yet not so securely that it cannot be disassembled, this is usually referred to as a “location” or “transition” fit. (middle)
- **Interference fits** for when it is desirable for the shaft to be securely held within the hole, this is usually referred to as a “press fit”, “friction fit”, or “interference fit”. (tight)

Holes

- Drilled holes will be oversized & out-of-round
 - OK for clearance holes
- Holes can be drilled 5% undersize and then reamed to a precision size (.001")
 - Reamers are available in standard sizes, and also oversize & undersize near the standard sizes
- You can make a hole to a non-standard diameter if you bore it because you can adjust the tool in the boring bar accurately to .001"
 - For this project, in most cases standard sizes shall do the job

Holes



Drills

Multi-purpose hole drilling
(pre-drilling, clearance
holes, etc.)

Hole precision of
0.004"-0.010" depending
on size

Used in ME 350 ✓



Reamers

For precision holes (close
or interference fits)

Hole precision of
0.0001"-0.0005"
depending on size

Used in ME 350 ✓

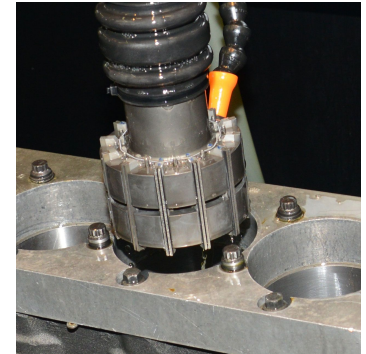


Boring bars

Larger precision holes
(large press fits/close fits)

Hole precision of 0.001"

Not used in ME 350 ✗



Honing Tools

Larger precision holes
(close fits)

Not used in ME 350 ✗

Tapping

Ex: To drill a free fit hole for 1/4-20 screw, what is the clearance hole drill?

To create a 1/4-20 tapped hole, what is the tap drill?

Table 4. Tap Drills and Clearance Drills for Machine Screws with American National Thread Form

Size of Screw		No. of Threads per Inch	Tap Drills		Clearance Hole Drills			
No. or Diam.	Decimal Equiv.		Drill Size	Decimal Equiv.	Close Fit		Free Fit	
					Drill Size	Decimal Equiv.	Drill Size	Decimal Equiv.
0	.060	80	3/64	.0469	52	.0635	50	.0700
1	.073	64 72	53 53	.0595 .0595	48	.0760	46	.0810
2	.086	56 64	50 50	.0700 .0700	43	.0890	41	.0960
3	.099	48 56	47 45	.0785 .0820	37	.1040	35	.1100
4	.112	36 ^a 40 48	44 43 42	.0860 .0890 .0935	32	.1160	30	.1285
5	.125	40 44	38 37	.1015 1040	30	.1285	29	.1360
6	.138	32 40	36 33	.1065 .1130	27	.1440	25	.1495
8	.164	32 36	29 29	.1360 .1360	18	.1695	16	.1770
10	.190	24 32	25 21	.1495 1590	9	.1960	7	.2010
12	.216	24 28	16 14	.1770 .1820	2	.2210	1	.2280
14	.242	20 ^a 24 ^a	10 7	.1935 .2010	D	.2460	F	.2570
1/4	.250	20 28	7 3	.2010 .2130	F	.2570	H	.2660
5/16	.3125	18 24	F I	.2570 .2720	P	.3230	Q	.3320



Tapping

Solution:

- Clearance hole drill:
 - Slip fit: 1/4" drill bit
 - Close fit: F drill
 - Free fit: H drill
- Tap drill: 1/4-20
 - #7 drill (Imperial)

Table 4. Tap Drills and Clearance Drills for Machine Screws with American National Thread Form

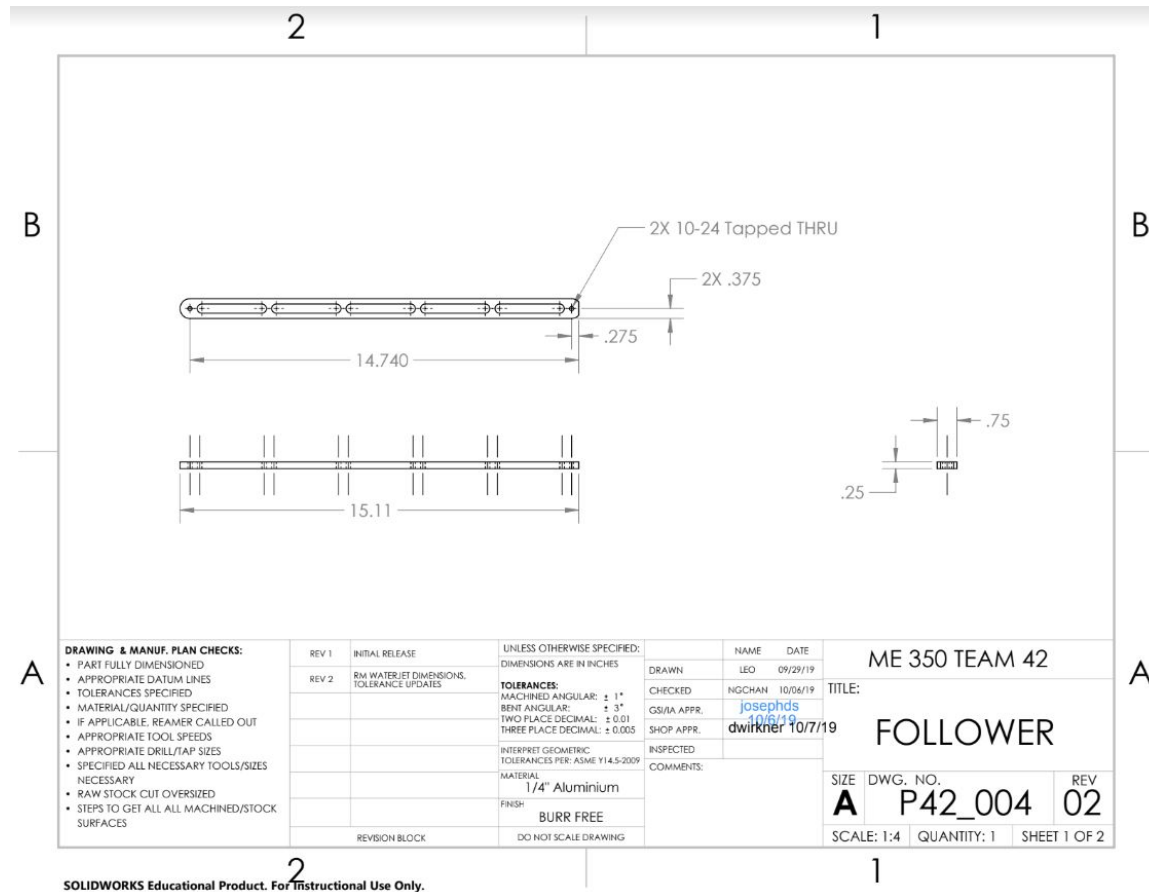
Size of Screw		No. of Threads per Inch	Tap Drills		Clearance Hole Drills			
No. or Diam.	Decimal Equiv.		Drill Size	Decimal Equiv.	Close Fit		Free Fit	
					Drill Size	Decimal Equiv.	Drill Size	Decimal Equiv.
0	.060	80	3/64	.0469	52	.0635	50	.0700
1	.073	64 72	53 53	.0595 .0595	48	.0760	46	.0810
2	.086	56 64	50 50	.0700 .0700	43	.0890	41	.0960
3	.099	48 56	47 45	.0785 .0820	37	.1040	35	.1100
4	.112	36 ^a 40 48	44 43 42	.0860 .0890 .0935	32	.1160	30	.1285
5	.125	40 44	38 37	.1015 1040	30	.1285	29	.1360
6	.138	32 40	36 33	.1065 .1130	27	.1440	25	.1495
8	.164	32 36	29 29	.1360 .1360	18	.1695	16	.1770
10	.190	24 32	25 21	.1495 1590	9	.1960	7	.2010
12	.216	24 28	16 14	.1770 .1820	2	.2210	1	.2280
14	.242	20 ^a 24 ^a	10 7	.1935 .2010	D	.2460	F	.2570
1/4	.250	20 28	7 5	.2010 .2130	F	.2570	H	.2660
5/16	.3125	18 24	F I	.2570 .2720	P	.3230	Q	.3320

Press Fits

- **Measure** the outer diameter of the bushing before drilling
 - Should be around .315", different from specs on McMaster.com
- **Pre-drill** undersized hole
 - Approximately .015" smaller than the finished hole diameter
 - Assuming a .315" bushing, pre-drill hole size should be ~.299" -> 19/64" drill bit
- **Ream** to final size
 - The final size should be .001" smaller than the measured diameter of the component to be pressed in
 - Assuming a .315" bushing, final size of the hole should be .314" -> .3140" reamer

Datums

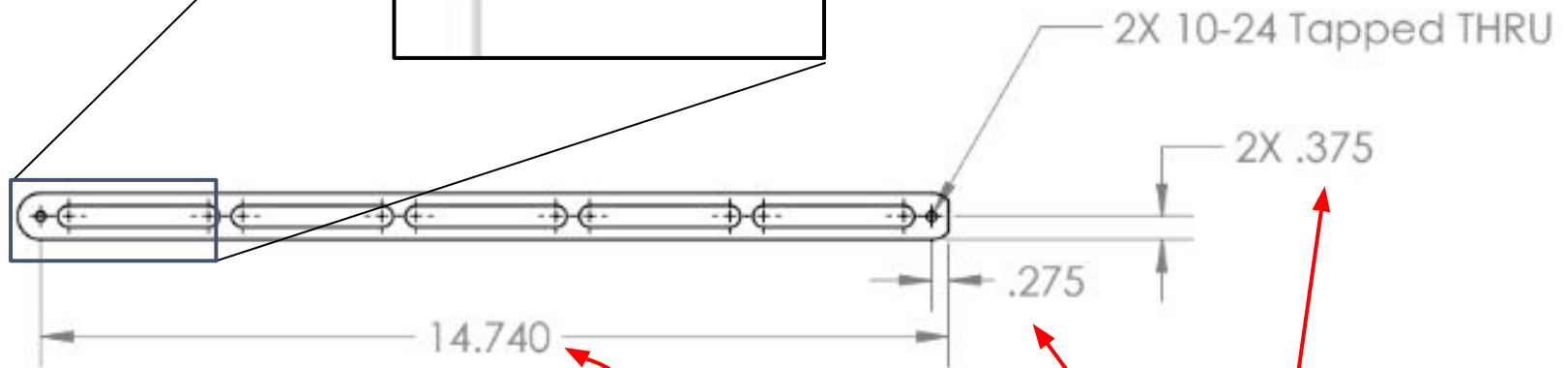
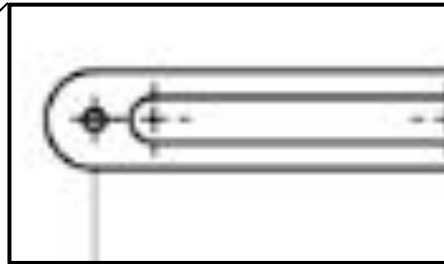
- Use a single datum for each drawing
 - Could be an edge
 - Could be a center of a hole
- DO NOT miss any dimension



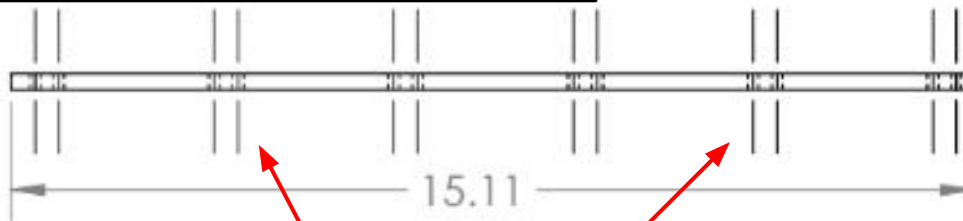
SOLIDWORKS Educational Product. For Instructional Use Only.

Datums

✓ Center lines included



✗ precision holes should be dimensioned relative to **each other**, not an edge



✗ unnecessary center lines

✓ 3-place decimal from datum for precision hole

✓ hidden lines show it is a through hole

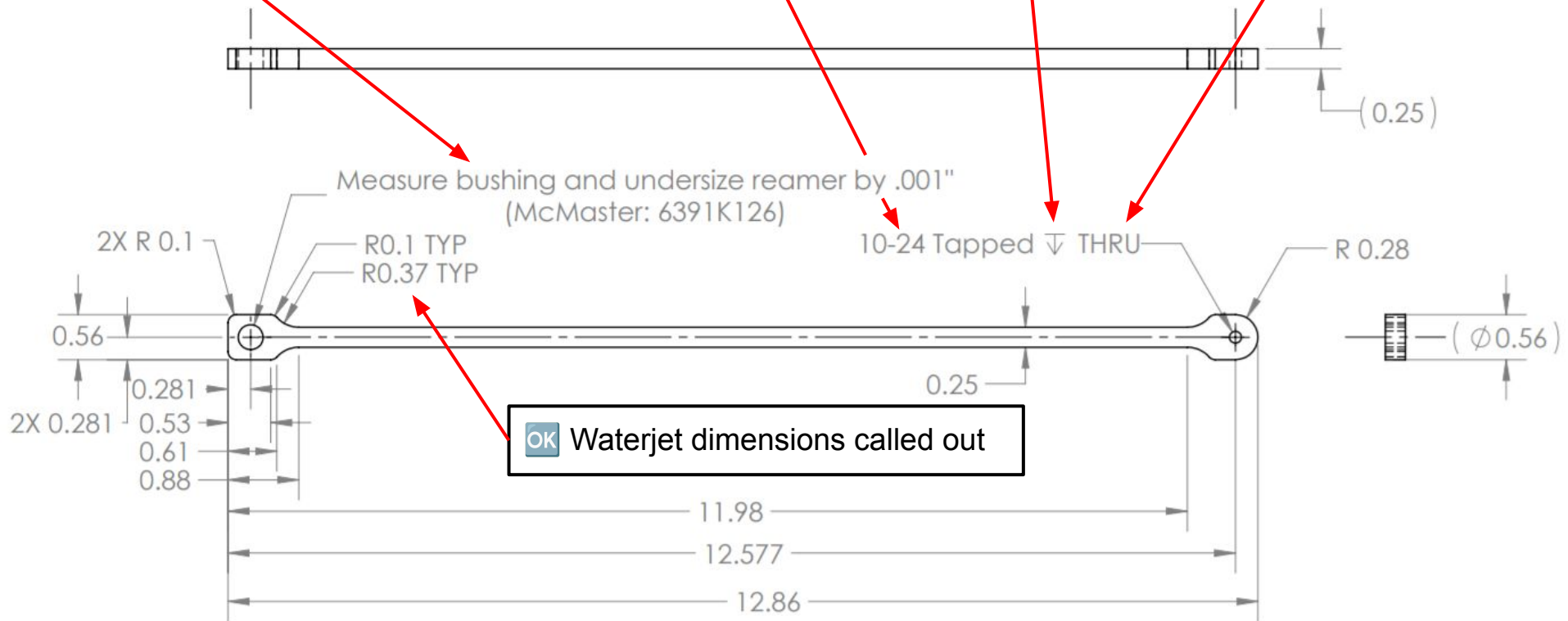
Hole Callouts

- ✓ Part number of part to be pressed in
- ✓ Instructions to measure specific part before machining feature

✓ Thread size
(Drill size given
in Mfg. Plan)

✓ Depth
symbol

✓ Depth
specification
(THRU)



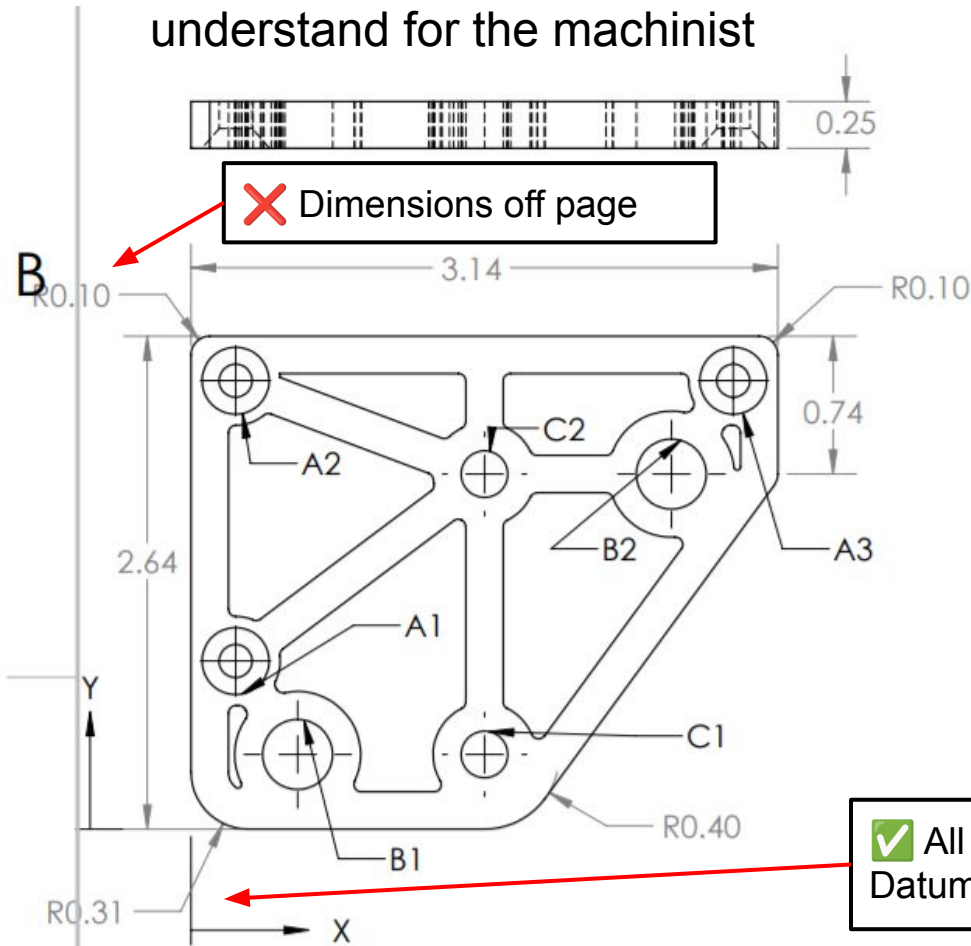
Hole Tables

Use a hole table when your part has a lot of holes - it is much cleaner and easier to understand for the machinist

✓ Hole ID/Tag

✓ X and Y locations

✓ Hole size and specification



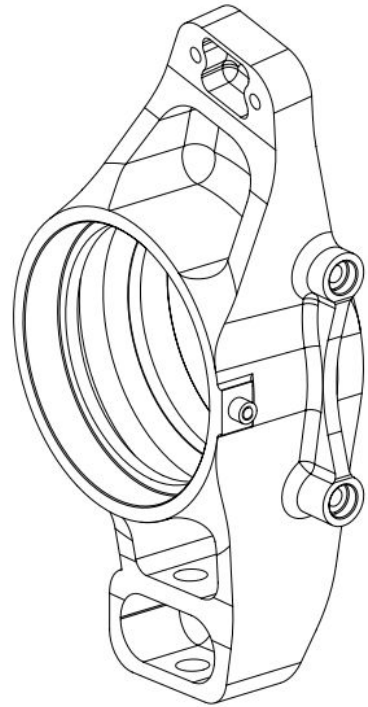
TAG	X LOC	Y LOC	SIZE
A1	.239	.900	ϕ 0.177 THRU ALL ✓ ∇ 0.112 X 82°
A2	.239	2.400	ϕ 0.177 THRU ALL ✓ ∇ 0.112 X 82°
A3	2.902	2.400	ϕ 0.177 THRU ALL ✓ ∇ 0.112 X 82°
B1	.570	.400	ϕ 0.375 THRU ALL
B2	2.570	1.900	ϕ 0.375 THRU ALL
C1	1.570	.400	SEE CALLOUT BELOW
C2	1.570	1.900	SEE CALLOUT BELOW

C1 and C2 are reamed for .001" interference fit with 0.250" dowel

✓ All holes relative to Datum X and Y

✓ Specific instructions for specific holes

- NOTES:
1. THIS IS A MINIMAL DIMENSION DRAWING
SEE CAD FILE FOR BASIC GEOMETRY
 2. THIS DRAWING IS FOR THE RIGHT SIDE PART
THE LEFT SIDE PART IS MIRRORED ABOUT DATUM B



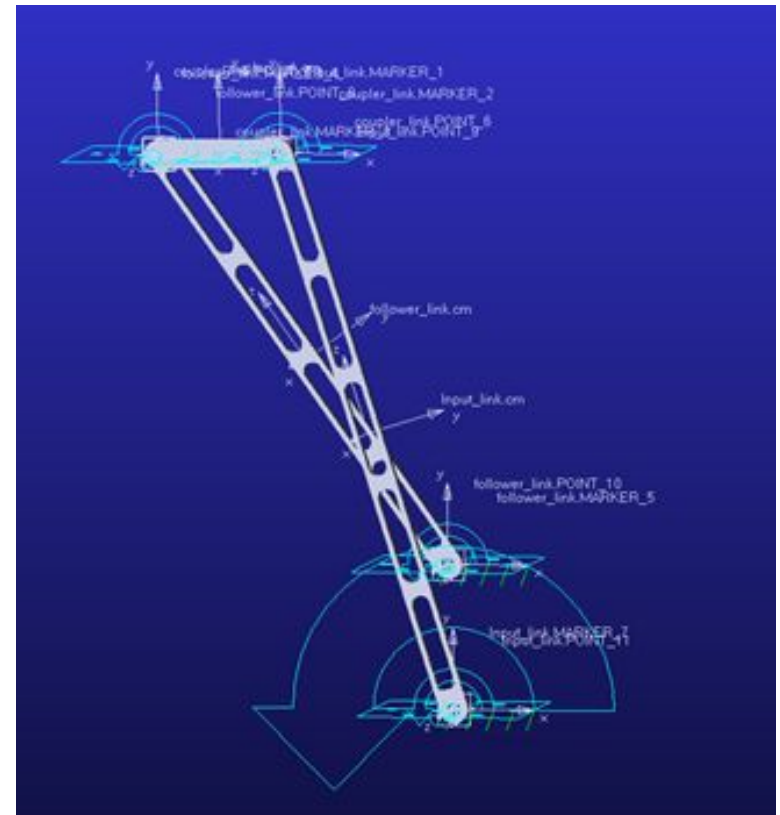
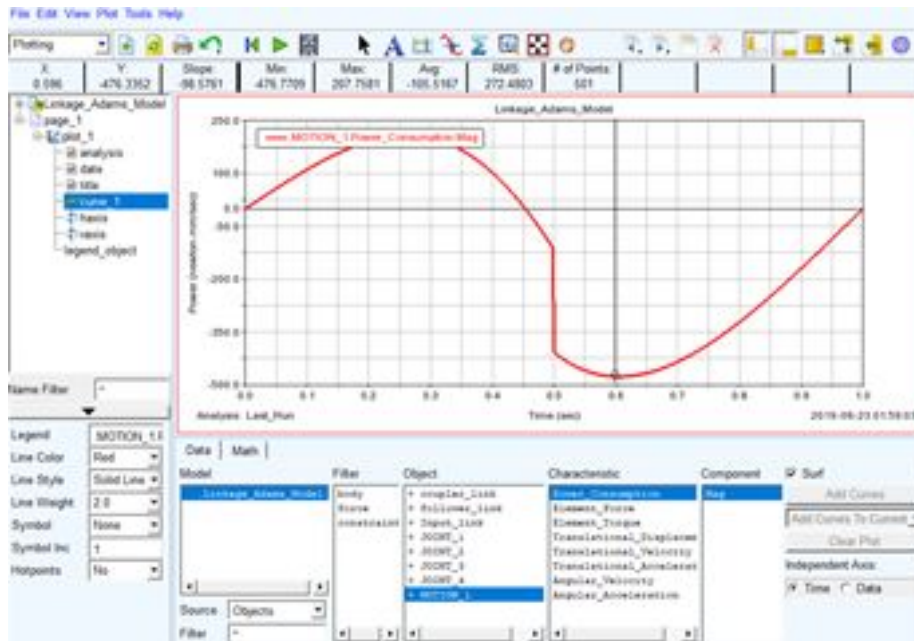
QUANTITY 1		THIRD ANGLE PROJECTION 		DECIMALS .X ± .1 .XX ± .01 .XXX ± .003		ANGLES X. ± 1° .X ± .1°		125 ✓		MATERIAL: 7075 - T6S1 Aluminum		MRacing 2018				TITLE: MRacing Front Upright Right	
						INTERPRET DRAWING PER ASME Y14.5M-2009 REMOVE BURRS BREAK ALL SHARP EDGES .005-.015		DESIGNED: Brendan Prast DRAWN: Brendan Prast		DATE: 1/8/18 DATE: 1/14/18		FILE NAME: MR18_Front_Upright _Right.stp		REVISION A		SHEET 1 of 3 SIZE B	
						DRAWING NOT TO SCALE DO NOT SCALE DRAWING		DATE: 1/8/18 DATE: 1/14/18		DATE: 1/8/18 DATE: 1/14/18							

ADAMS Tutorial 3+4



Today

- Run through Lab 3 (ADAMS Tutorial Parts 3 & 4)
 - Goal: Create a dynamic simulation of your individual model



ADAMS Pro Tips

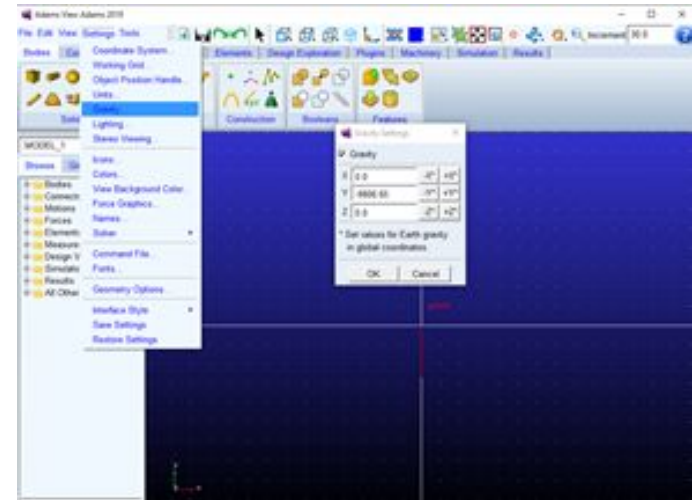
- Assign densities, then start at part 4
- Move your linkage to position 1 before saving as a Parasolid (.x_t)
- Measure angle between position 3 and 1 in SolidWorks to get total travel
- Delete all bodies not associated with your linkage (ex: playing field) or make a new assembly just for ADAMS
- Common material densities: **Enter like this: (2700 (kg/m**3))**
 - Aluminum: 2700 kg/m³
 - Steel: 8050 kg/m³
 - ABS Plastic: 900 - 1530 kg/m³
 - Acrylic: 1180 kg/m³
 - Flashlight: ??? kg/m³



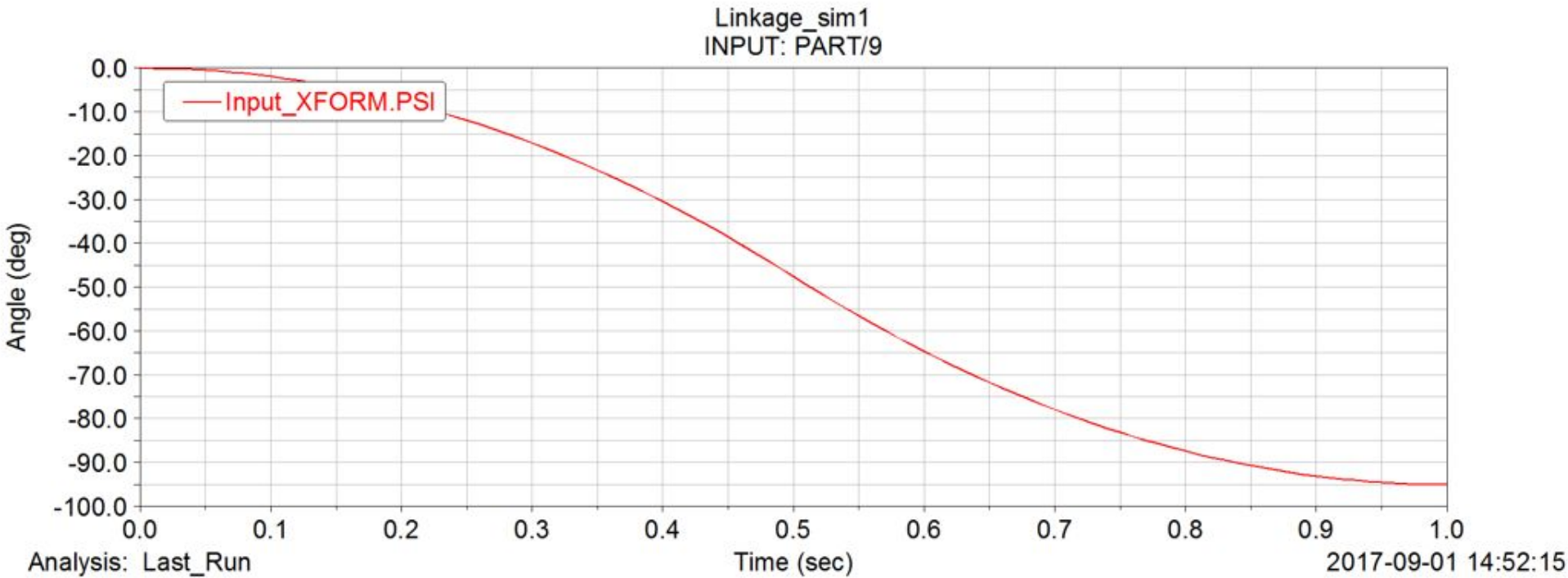
ADAMS Pro Tips

- Run your simulations with No Gravity, or make sure it is acting perpendicular to the plane of your linkage
- To change gravity, go to Settings > Gravity (see right)
- Make sure to change “displacement” to “Acceleration” for the motor motion
- Motor Specs:

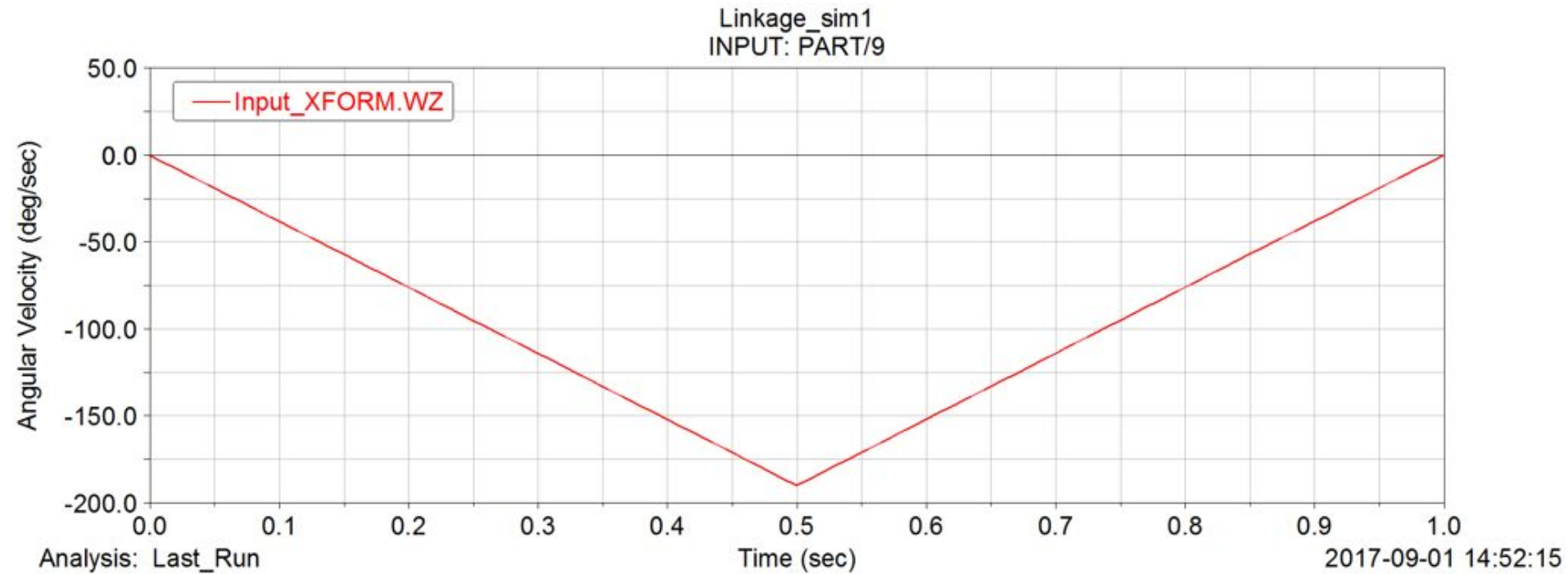
<https://www.pololu.com/product/2823/specs>



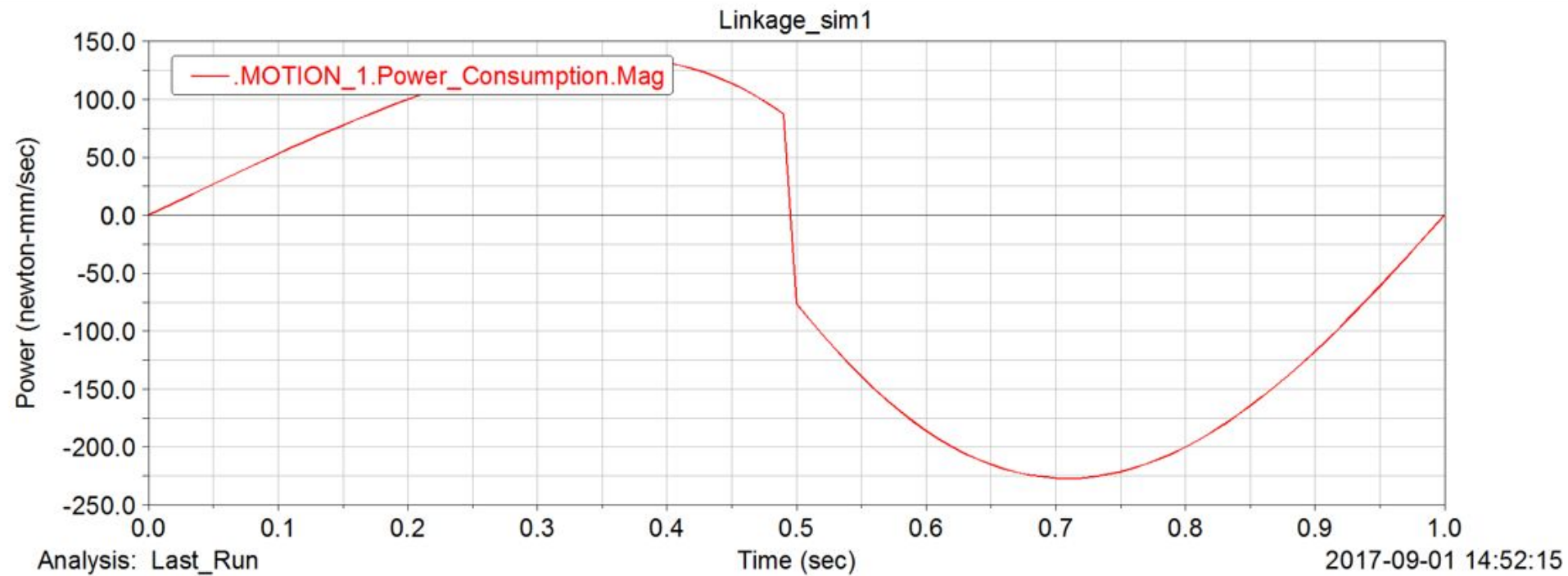
ADAMS Plot Example



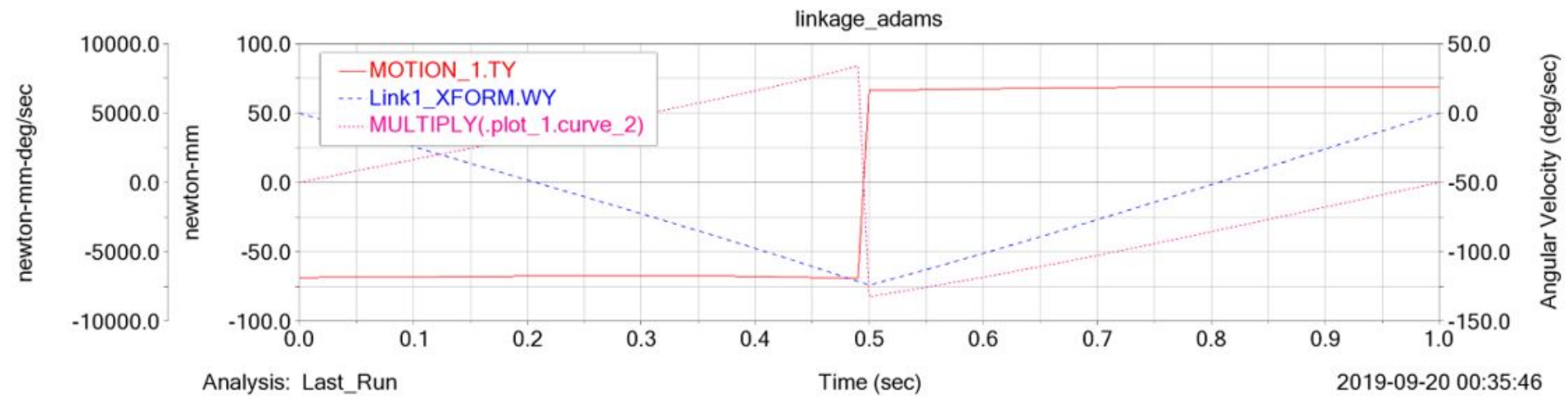
ADAMS Plot Example



ADAMS Plot Example



ADAMS Plot Example



Summary

Run through Lab 3 ([ADAMS Tutorial Parts 3 & 4](#))

- Goal: Create a dynamic simulation of your individual model

Due next lab:

1. [Lab Assignment 3 Worksheet](#) (Individual Assignment)
 - a. Screenshots from Individual ADAMS model
2. Gate Review #1 (Team Assignment)
 - a. Schedule with GSI/IA for any time between 1/X and the day before Lab 4
 - b. See [Gate Review #1 Expectations](#)
3. Manufacturing Review (Team Assignment)
 - a. Start on this early, but also schedule your Gate Review #1 so that you have time to address GSI/IA concerns before presenting your manufacturing review to the class
 - b. See [Manufacturing Review Expectations](#)

